

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Viva Based Questions

Course Code:	DSA0404	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO4	Assessment:	Assessment Tool 3– Viva Based Questions
Weightage:	20% of CIA	Total Marks:	100
CO3 Statement:	To statistically infer Viva Based Questions		
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Code of Conduct

I M Poojitha Reddy - 192472352 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: M Poojitha Reddy

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

A. Basics of Statistical Inference

1. What is statistical inference?

Statistical inference is the process of using **sample data to draw conclusions or make predictions about a larger population**. It helps estimate unknown population parameters and test claims or hypotheses.

Example: A company surveys 500 customers and uses their responses to estimate the satisfaction level of all customers.

2. Why is statistical inference important in data science?

Statistical inference is important because data scientists usually cannot collect data from an entire population. It helps them:

- Estimate population characteristics from samples.
 - Test assumptions and claims.
 - Measure uncertainty in predictions.
 - Make data-driven decisions.
 - Determine whether observed differences are statistically significant.
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3. What is the difference between descriptive statistics and inferential statistics?

Descriptive Statistics	Inferential Statistics
Summarizes collected data	Draws conclusions about a population
Uses mean, median, graphs, etc.	Uses estimation, confidence intervals, hypothesis tests
Describes what the data shows	Predicts or generalizes beyond the data
Example: Average salary of 500 employees	Estimating average salary of all employees

4. What is a population in statistics?

A **population** is the complete set of individuals, objects, or observations about which a statistical study is concerned.

Example: All customers of an online shopping company.

5. What is a sample?

A **sample** is a smaller subset selected from a population for analysis.

Example: 1,000 randomly selected customers from all customers of an online shopping company.

6. Why do data scientists usually work with samples instead of complete populations?

Working with the complete population may be:

- Expensive
- Time-consuming
- Difficult or impossible
- Resource-intensive

Therefore, data scientists collect a representative sample and use statistical inference to draw conclusions about the population.

7. What is a population parameter?

A **population parameter** is a numerical value that describes a characteristic of the entire population.

Examples:

- Population mean = μ
- Population proportion = p
- Population variance = σ^2
- Population standard deviation = σ

Usually, population parameters are unknown.

8. What is a sample statistic?

A **sample statistic** is a numerical value calculated from sample data.

Examples:

- Sample mean = $\bar{x}$
- Sample proportion = $\hat{p}$
- Sample variance = s^2
- Sample standard deviation = s

Statistics are used to estimate population parameters.

9. Give a real-world example where statistical inference is used in industry.

A pharmaceutical company tests a new medicine on a sample of 1,000 patients. The company uses the sample results to estimate the effectiveness and safety of the medicine for the larger patient population.

10. What are the two major activities involved in statistical inference?

The two major activities are:

1. **Estimation** — estimating unknown population parameters using sample data.
2. **Hypothesis Testing** — testing claims or assumptions about population parameters.

Inference → Estimation + Hypothesis Testing

B. Frequentist Approach

11. What is the frequentist approach to statistical inference?

The **frequentist approach** is a method of statistical inference where probabilities are interpreted in terms of the **long-run frequency of events** over repeated sampling.

The population parameter is considered fixed, while the sample data is considered random.

12. What is the basic idea behind frequentist statistics?

The basic idea is:

Use repeated random samples and their behavior to make conclusions about fixed but unknown population parameters.

For example, a sample mean varies from sample to sample, while the true population mean remains fixed.

13. How does the frequentist approach differ from the Bayesian approach?

Frequentist	Bayesian
Parameter is fixed	Parameter is treated as uncertain/random
Data is random	Data is observed
Uses sampling distributions	Uses prior + likelihood
Uses p-values and confidence intervals	Uses posterior probabilities and credible intervals
Does not require a prior distribution	Uses prior information

14. In the frequentist approach, what is considered fixed: the parameter or the data?

In the frequentist approach:

Population parameter = Fixed

Sample data = Random

The parameter is unknown but is assumed to have one fixed true value.

15. What does long-run frequency mean in frequentist statistics?

Long-run frequency refers to the proportion of times an event occurs if an experiment is repeated **many times under the same conditions**.

For example, if a fair coin is tossed millions of times, the proportion of heads will approach 50%.

16. Why is random sampling important in the frequentist approach?

Random sampling ensures that every population member has a known chance of being selected. It helps:

- Reduce selection bias.
 - Produce representative samples.
 - Establish valid sampling distributions.
 - Make reliable population inferences.
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17. Can a population parameter be considered a random variable in the frequentist approach? Why?

No.

In the frequentist approach, the population parameter is considered a **fixed but unknown quantity**.

For example, the true population mean μ has one fixed value. We do not assign a probability distribution to μ .

18. Give an example of a frequentist problem in a data science application.

A data scientist wants to determine whether a new machine-learning model has higher accuracy than an existing model.

They:

1. Select a random test sample.
2. Calculate both model accuracies.
3. Formulate H_0 and H_1 .
4. Calculate a test statistic and p-value.
5. Decide whether the improvement is statistically significant.

This is a frequentist hypothesis-testing problem.

C. Point Estimation

19. What is a point estimate?

A **point estimate** is a single numerical value used to estimate an unknown population parameter.

Example:

If the sample mean is 72 marks, then:

$$\bar{x} = 72$$

is the point estimate of the population mean μ .

20. What is an estimator?

An **estimator** is a rule or formula used to estimate a population parameter from sample data.

For estimating the population mean:

$$\text{Estimator} = \bar{x} = \Sigma x/n$$

The estimator is a random quantity because it depends on the random sample.

21. What is the difference between an estimator and an estimate?

Estimator	Estimate
Formula or rule used for estimation	Actual numerical result
Example: $\bar{x}$	Example: $\bar{x} = 72$
Random variable	Fixed value after sample is observed

22. What is the point estimator for the population mean?

The sample mean is the point estimator of the population mean.

$$\bar{x} = \Sigma x_i/n$$

where:

- x_i = individual observations
 - n = sample size
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23. How do you estimate a population proportion from a sample?

The population proportion p is estimated using the sample proportion:

$$\hat{p} = x/n$$

where:

- x = number of successes
- n = sample size

Example: If 80 out of 100 customers are satisfied:

$$\hat{p} = 80/100 = 0.80 = 80\%$$

24. What is the point estimate of population variance?

The sample variance is commonly used as the unbiased point estimator of population variance.

$$s^2 = \Sigma(x_i - \bar{x})^2 / (n - 1)$$

The denominator $n - 1$ provides an unbiased estimator of the population variance.

25. Why is the sample mean commonly used to estimate the population mean?

The sample mean is commonly used because it is:

- **Unbiased**
- Consistent
- Easy to calculate
- Efficient under common assumptions
- Based on all observations in the sample

Therefore, it is a reliable estimator of the population mean.

26. What properties should a good estimator have?

A good estimator should ideally have:

1. **Unbiasedness** — its expected value equals the true parameter.
 2. **Consistency** — approaches the true parameter as sample size increases.
 3. **Efficiency** — has relatively low variance.
 4. **Reliability** — produces stable estimates from samples.
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27. What is an unbiased estimator?

An estimator is **unbiased** if its expected value is equal to the true population parameter.

Mathematically:

$$E(\hat{\theta}) = \theta$$

For example, the sample mean is an unbiased estimator of the population mean:

$$E(\bar{x}) = \mu$$

28. What is meant by the bias of an estimator?

Bias is the difference between the **expected value of an estimator and the true population parameter**.

$$\text{Bias}(\hat{\theta}) = E(\hat{\theta}) - \theta$$

If:

$$\text{Bias} = 0$$

the estimator is unbiased.

29. What is the difference between bias and variance of an estimator?

Bias	Variance
Measures systematic error	Measures variability

Bias	Variance
Difference between expected estimate and true value	Variation of estimates across samples
Low bias means estimates are centered around true value	Low variance means estimates are more consistent

A good estimator generally aims for **low bias and low variance**.

30. Why can two different samples produce different point estimates for the same population parameter?

Different random samples contain different observations. Therefore, their statistics such as sample mean or sample proportion may differ.

This is called **sampling variability**.

For example:

Sample 1 mean = 70

Sample 2 mean = 73

Both can be valid estimates of the same population mean.

D. Variability in Estimates

31. Why does a point estimate have uncertainty?

A point estimate is calculated from a sample, and different samples can produce different estimates.

Therefore, the estimate may not exactly equal the true population parameter.

This uncertainty is called **sampling uncertainty**.

32. What is the sampling distribution of an estimator?

The **sampling distribution** is the probability distribution of an estimator obtained by considering **all possible random samples of a fixed size** from a population.

For example, the sampling distribution of $\bar{x}$ describes how sample means vary across repeated samples.

33. What is the standard error?

The **standard error (SE)** measures the variability of a sample statistic or estimator across repeated samples.

For the sample mean:

$$SE(\bar{x}) = \sigma/\sqrt{n}$$

If σ is unknown, it is commonly estimated as:

$$SE(\bar{x}) = s/\sqrt{n}$$

34. How is standard error different from standard deviation?

Standard Deviation	Standard Error
Measures variability of individual observations	Measures variability of an estimator
Describes spread within a sample/population	Describes sampling variability
Usually does not automatically decrease with n	Generally decreases as sample size increases
Example: spread of individual salaries	Uncertainty in estimated average salary

35. How does increasing the sample size affect the standard error of the sample mean?

The standard error is:

$$SE = \sigma/\sqrt{n}$$

As sample size **n increases**, $\sqrt{n}$ increases, so the standard error **decreases**.

For example:

$$n = 100 \rightarrow SE = \sigma/10$$

$$n = 400 \rightarrow SE = \sigma/20$$

Thus, larger samples produce more precise estimates.

36. Why does variability in an estimate decrease when the sample size increases?

A larger sample contains more observations, so individual random fluctuations have less influence on the overall estimate.

Mathematically:

$$SE \propto 1/\sqrt{n}$$

Therefore, increasing sample size reduces sampling variability.

37. What is the relationship between standard error and confidence intervals?

The standard error measures the uncertainty of an estimate, while the confidence interval uses the standard error to construct a range around the estimate.

General form:

$$\text{Confidence Interval} = \text{Estimate} \pm \text{Critical Value} \times SE$$

Therefore:

Larger SE → Wider CI

Smaller SE → Narrower CI

38. If two samples have different sample means, does it necessarily mean that one sample is incorrect? Explain.

No.

Different random samples can naturally produce different sample means due to **sampling variability**.

Both samples can be correct and valid. Statistical inference helps determine whether the observed difference is simply due to random variation or represents a meaningful population difference.

E. Confidence Intervals

39. What is a confidence interval?

A **confidence interval (CI)** is a range of plausible values for an unknown population parameter, calculated from sample data at a specified confidence level.

General form:

$$\text{CI} = \text{Estimate} \pm \text{Margin of Error}$$

Example:

$$95\% \text{ CI} = (45, 52)$$

This gives a range of plausible values for the population parameter.

40. Why is a confidence interval more informative than a point estimate?

A point estimate gives only one value, while a confidence interval provides:

- The estimated value.
- The uncertainty around the estimate.
- A range of plausible population values.
- An indication of precision.

Example:

Point estimate = ₹50,000

95% CI = ₹48,000 to ₹52,000

The interval gives much more information about uncertainty.

41. What does a 95% confidence level mean in the frequentist approach?

A 95% confidence level means that if we repeatedly take random samples and construct confidence intervals using the same method, approximately **95% of those intervals would contain the true population parameter**.

It does **not** strictly mean that there is a 95% probability that the already-calculated interval contains the parameter.

42. What is the difference between a confidence level and a confidence interval?

Confidence Level	Confidence Interval
Percentage such as 90%, 95%, 99%	Actual numerical range
Describes the long-run reliability of the method	Gives plausible values for the parameter
Example: 95% confidence	Example: (45, 52)

43. What factors determine the width of a confidence interval?

The main factors are:

1. **Sample size**
2. **Variability/standard error**
3. **Confidence level**
4. **Critical value**

General relationship:

Width $\propto$ Critical Value $\times$ Standard Error

44. How does increasing the sample size affect the width of a confidence interval?

Increasing the sample size decreases the standard error:

$$SE = \sigma / \sqrt{n}$$

Therefore, the margin of error decreases and the confidence interval becomes **narrower**.

Thus:

Larger sample $\rightarrow$ Smaller SE $\rightarrow$ Narrower CI $\rightarrow$ Greater precision

45. How does increasing the confidence level from 95% to 99% affect the confidence interval?

Increasing the confidence level increases the critical value.

Therefore, the margin of error increases and the confidence interval becomes **wider**.

95% CI → Narrower

99% CI → Wider

The 99% interval provides greater confidence but less precision.

46. What is the difference between a z-confidence interval and a t-confidence interval?

z-Confidence Interval

Uses normal distribution

Population σ is known or conditions justify z

Common for large samples

Critical values come from z-distribution

t-Confidence Interval

Uses t-distribution

Population σ is unknown

Especially useful for small samples

Critical values depend on degrees of freedom

47. When would you use the t-distribution instead of the normal distribution for estimating a population mean?

The **t-distribution** is generally used when:

- Population standard deviation **σ is unknown**.
- Sample standard deviation **s** is used instead.
- Especially when the sample size is small.
- The population is reasonably approximately normal for small samples.

For a one-sample mean:

$$t = (\bar{x} - \mu) / (s / \sqrt{n})$$

with:

$$df = n - 1$$

F. Hypothesis Testing Using Confidence Intervals

48. What is hypothesis testing?

Hypothesis testing is a statistical method used to determine whether sample data provides sufficient evidence to support or reject a claim about a population parameter.

Basic process:

Claim → **H₀/H₁** → **Test Statistic** → **p-value** → **Decision** → **Conclusion**

49. What are the null hypothesis (H₀) and alternative hypothesis (H₁)?

Null Hypothesis (H₀):

The null hypothesis represents the **default assumption**, usually stating that there is no difference, no effect, or that a parameter equals a specified value.

Alternative Hypothesis (H₁):

The alternative hypothesis represents the claim or effect that we want to find evidence for.

Example:

A company claims average response time is 2 seconds.

H₀: $\mu = 2$

H₁: $\mu < 2$

The hypotheses depend on the research question and whether the test is left-tailed, right-tailed, or two-tailed.

50. How can a confidence interval be used to make a decision about a hypothesis?

A confidence interval can be compared with the value stated in the null hypothesis.

For example, suppose:

$H_0: \mu = 50$

and the 95% CI is:

(53, 58)

Since **50 is not inside the confidence interval**, we reject H_0 at the corresponding 5% significance level.

Decision Rule

Confidence Interval	Decision
Claimed value is inside CI	Fail to reject H_0
Claimed value is outside CI	Reject H_0